Milestone 2 Report

ASCEND: Adaptive Self-supervised Continual Expert Network for Domain-Incremental Wi-Fi Sensing

Course	ECE 310 Wireless Communication, Monsoon Semester 2026
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Programme	B.Tech Computer Science Engineering
Domain	Wi-Fi Sensing using CSI for Human Activity Recognition
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1. Problem Statement and Milestone Summary

Wi-Fi sensing uses Channel State Information (CSI) to recognise human activities. It needs no camera and no wearable device. But a model trained in one room often fails in another. Walls, furniture and device positions all change the signal.

This work studies **domain-incremental Wi-Fi Human Activity Recognition (HAR)**. New environments arrive one after another. The model must learn each one. It must also keep what it learned before. When it forgets old environments, this is called **catastrophic forgetting**. Labelled data is also limited. Storing old samples brings privacy and storage problems.

The central question is: *Can a Wi-Fi sensing model learn new environments continuously, without forgetting old ones, through self-supervised learning?*

Milestone 1 set the problem, reviewed the literature and found the research gap. Milestone 2 reports four things:

- The ASCEND design and its five improvements over the closest existing work, SAMoE-C.
- The experiment setup on the MM-Fi dataset with four rooms in sequence.
- Preliminary results against baselines and SAMoE-C.
- The plan for the final two weeks.

2. Review of Existing Work

I studied four main papers. They cover continual Wi-Fi sensing, cross-domain HAR, self-supervised Wi-Fi sensing and mixture-of-experts (MoE) models. The table shows what each one does.

Work	Main idea	SSL	Continual	Forgetting handled	Relevance to ASCEND
Wi-DMAR [2] (2026)	Conditional diffusion for cross-domain HAR	No	No	No	Deals with domain shift. Not continual.
CIG-MAE [3] (2026)	Cross-modal masked autoencoder	Yes	No	No	Shows learning with fewer labels.
ConSense [1] (2025)	Continual sensing by growing and picking	No	Yes	Yes	Shows why continual learning is needed.
SAMoE-C [4] (2026)	Scene-adaptive continual learning with MoE	No	Yes	Yes	Closest work. Base paper for ASCEND.
ASCEND (proposed)	Self-supervised continual expert network	**Yes**	**Yes**	**Yes**	Has all three properties.

Figure 1 shows how the two closest papers work.

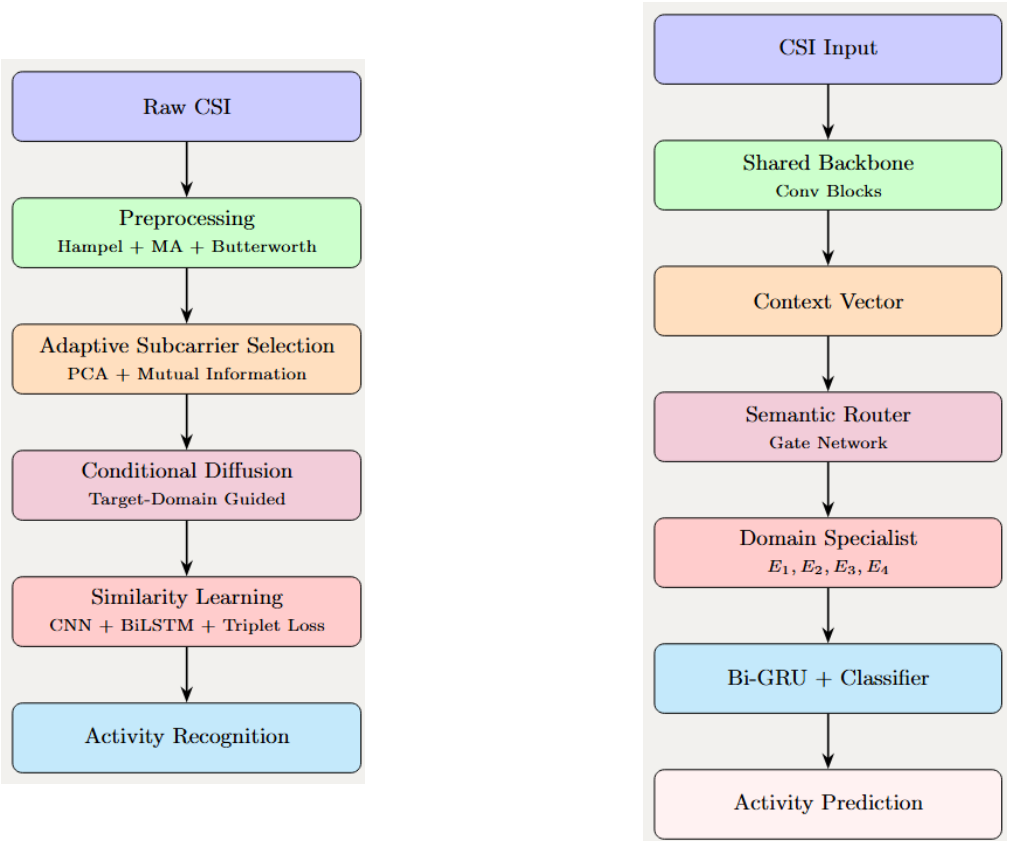


Fig. 1: Left: Wi-DMAR [2]. Right: SAMoE-C [4].

Key observations

- Catastrophic forgetting affects every continual setup.
- Labels are the main bottleneck in Wi-Fi sensing.
- Keeping old samples can risk privacy and needs storage.
- Continual domain adaptation for Wi-Fi HAR is still largely unexplored.

3. Current SOTA Position and Research Direction

SAMoE-C [4] is the closest work. It has a frozen shared backbone, a gate-based router and one specialist per domain. It handles continual learning and forgetting. But it trains with labels only. ASCEND keeps the expert idea. It adds self-supervised learning and five improvements. The table compares the two.

Component	SAMoE-C [4]	ASCEND (proposed)	Expected benefit
EWC	Backbone frozen	Backbone adapts using EWC	Less forgetting across domains
SSL Router	Gate only	Gate plus prototype	Better choice of expert
SSL-KD	Label supervision only	Backbone teaches specialists	Better aligned features
AIM Masking	No masking	Learned masking policy	Stronger self-supervised signal
Balanced Replay	Replay biased to newest domain	Equal replay from all domains	Less bias toward recent domains

Positioning. No paper I reviewed combines four things: Wi-Fi CSI sensing, domain-incremental learning, self-supervised learning and forgetting control. ASCEND aims to fill this gap. Section 6 gives first evidence against SAMoE-C.

4. Proposed Method: ASCEND

ASCEND works in three phases, shown in Figure 2. The input is a CSI tensor X.

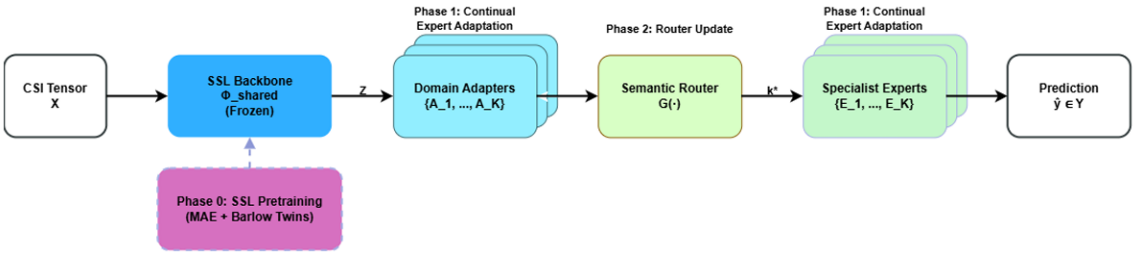


Fig. 2: ASCEND workflow. Phase 0 trains the backbone without labels. Phase 1 adds an adapter and a specialist for each domain. Phase 2 updates the router.

- **Phase 0: SSL pretraining.** The backbone learns from unlabelled CSI. It uses a Masked Autoencoder (MAE) and Barlow Twins together. MAE hides parts of the signal and rebuilds them. Barlow Twins keeps features consistent across two views of the same signal. The result is a shared representation Z.
- **Phase 1: Continual expert adaptation.** Each new domain gets a small adapter and a specialist expert. Older experts are left untouched. This limits interference between domains.
- **Phase 2: Router update.** The semantic router G picks the best expert, called k*. That expert predicts the activity.

The five improvements

- **EWC (Elastic Weight Consolidation):** protects weights that mattered for earlier domains.
- **SSL Router:** uses a gate and domain prototypes to choose the expert.
- **SSL-KD:** the backbone teaches the specialists, so their features stay aligned.
- **AIM Masking:** a learned policy hides the most informative parts of the signal during training.
- **Balanced Replay:** replays an equal share of old samples from every domain. This removes bias toward the newest domain.

Model variants

I also tested two lighter versions, ASCEND-Lite and ASCEND-ULite. They are meant for devices with limited resources.

5. Experimental Setup

I use the **MM-Fi** dataset [5]. The four domains are four rooms: two living rooms and two meeting rooms. They arrive in a fixed order, D1 to D4. The model trains on one room at a time.

Dataset item	Value
CSI shape	3 x 10 x 114 (antennas x time x subcarriers)
Activity classes	27
Subjects	40 humans
Domains	4 rooms (2 living + 2 meeting), in the order D1 to D4

Hyperparameter	Value	Hyperparameter	Value
CSI input	3 x 10 x 114	Mask ratio	0.75
Activities	27	EWC (lambda, gamma)	(1000, 0.90)
Domains	D1 to D4	Replay ratio	0.15
SSL epochs	100 / 50	KD weight	0.3
Learning rate	3×10^{-4}	Barlow Twins weight	0.2
Batch size	32		

Metrics. The main metric is accuracy (%) on each domain after the full sequence D1 to D4. I also report the average (Avg.). Accuracy on D1 to D3 after training on D4 shows how much the model kept.

6. Preliminary Results

All results here are **preliminary**. Full runs, repeated runs and ablations are planned for the last two weeks.

6.1 Sanity check on synthetic data

A first run on synthetic data checks that the pipeline works and that forgetting stays low.

Stage	D1	D2	D3
After D2	96.2	99.4	NA
After D3	91.2	96.2	100.0

After D3 is learned, accuracy on D1 falls from 96.2 to 91.2. That is a drop of 5.0 points. Forgetting is small.

6.2 Comparison on MM-Fi

Final accuracy (%) after training on D1 to D4 in order. Bold marks the best value in each column among the continual frameworks.

Method	D1	D2	D3	D4	Avg.
Monolithic baselines (sequential fine-tuning)					
AutoFi	3.46	4.25	4.85	67.00	19.89
ResNet-18	4.10	5.14	7.41	82.54	24.80
CAPC	3.60	4.17	4.16	57.89	17.45
MaskCAE	4.38	4.50	4.18	55.14	17.05
EWC	66.60	56.74	49.98	57.56	57.72
Continual learning frameworks					
SAMoE-C [4]	90.77	83.32	77.82	**88.23**	85.04
ASCEND	**94.21**	**89.96**	**89.13**	87.92	**90.30**
ASCEND-Lite	91.28	83.74	76.63	80.68	83.08
ASCEND-ULite	85.80	79.81	78.12	76.85	80.14

Citations for the baselines will be added in the final manuscript.

6.3 Analysis

- **Plain fine-tuning forgets almost everything.** AutoFi, ResNet-18, CAPC and MaskCAE score only 4 to 7.5% on D1 to D3. Their averages are 17.05 to 24.80%. They only do well on D4, the last domain.
- **EWC alone helps.** It keeps 49.98 to 66.60% on D1 to D3. Its average is 57.72%. This is why I use EWC inside ASCEND.
- **ASCEND beats SAMoE-C on average.** It scores 90.30% against 85.04%. That is a gain of 5.26 points.
- **The gain is biggest on older domains.** ASCEND is ahead by 3.44 points on D1, 6.64 on D2 and 11.31 on D3. This fits the goal of reducing forgetting.
- **SAMoE-C is slightly better on D4.** It scores 88.23% against 87.92%. The gap is 0.31 points. ASCEND gives up a little on the newest room to keep more on older rooms. Repeated runs will check this.
- **Lighter versions.** ASCEND-Lite (83.08%) is close to SAMoE-C (85.04%). ASCEND-ULite (80.14%) is lower.

7. Progress Completed

Area	Status	Progress
Domain understanding and problem formulation	Completed	Domain-incremental Wi-Fi HAR with limited labels and forgetting.
Literature review and gap	Completed	ConSense, Wi-DMAR, CIG-MAE and SAMoE-C compared on SSL, continual learning and forgetting.
Baseline results	Completed (preliminary)	Fine-tuning baselines, EWC and SAMoE-C results collected for MM-Fi in the order D1 to D4.
ASCEND implementation	First version done	Phases 0, 1 and 2 with all five improvements. Lite and ULite versions.
Preliminary experiments	Completed	Synthetic sanity check and MM-Fi comparison.
Full KPIs and ablations	In progress	To be finished in the final two weeks.
Manuscript	Started	Full draft will be written alongside the experiments.

8. Timeline and Next Milestones

Phase	Planned work	Status
Phase 1 (Weeks 1 to 3)	Domain understanding, problem formulation, literature study.	Completed
Phase 2 (Weeks 4 to 6)	Baseline reproduction, KPI planning, ASCEND coding.	Completed
Phase 3 (Weeks 7 to 9)	Experiments, KPI generation, first results.	Preliminary results done. Final runs in progress.
Phase 4 (Week 10)	Final paper and final presentation.	Planned

Next two weeks

- **Extract all KPIs:** accuracy per domain, average accuracy, forgetting measures and model size.
- **Run the ablation study:** remove EWC, the SSL Router, SSL-KD, AIM Masking and Balanced Replay one at a time.
- **Draft the full manuscript** in parallel.
- Repeat runs and try other domain orders.
- Test label efficiency with a fraction of the labels.

9. Limitations and Risks

- Results come from one domain order (D1 to D4). Variation across runs and orders is not yet measured.
- The effect of each of the five improvements is not yet separated. The ablation study will do this.
- Balanced Replay still keeps a small share of old samples (replay ratio 0.15). The paper must discuss the privacy and storage cost.
- The synthetic run is only a pipeline check. It does not show real-world performance.

10. Conclusion

Milestone 2 moved the project from a plan to a working system. ASCEND combines self-supervised pretraining, domain adapters, specialist experts and a router. It adds EWC, SSL-KD, AIM masking and balanced replay. On MM-Fi, the preliminary average accuracy is 90.30%. SAMoE-C scores 85.04%. Fine-tuning and EWC baselines score between 17 and 58%. The biggest gains are on older rooms. This supports the goal of less forgetting. The final two weeks will confirm these results with full KPIs and ablations. The manuscript will also be completed.

References

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